

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Given point P (-2, 6, 3) and vector $A = y \overline{ax} + (x + z) \overline{ay}$, Express P and A in Cylindrical and spherical coordinates. Evaluate A at P in Cartesian, cylindrical, and spherical systems.	Understand	6
B) Given $\overline{A} = 25 \overline{a\rho} + 12 \overline{a\phi} - 20 \overline{az}$ at $(8, 120^\circ, 5)$ find the vector component of $\overline{A}$ (i) Perpendicular to cylinder $\rho = 8$ (ii) tangent to the cylinder $\rho = 8$ (iii) tangent to the plane $\phi = 120^\circ$ (iv) a unit vector perpendicular to $\overline{A}$ and also tangent to $\rho = 8$	Understand	6
C) State and Verify Divergence Theorem		6
Q.2 Solve Any Two of the following.		12
A) Derive an Expression for Electric Field Intensity at any point due to infinite Line charge with charge density ρ_l C/m	Remember	6
B) A Line charge density $\rho_l = 15nC/m$ is located in free space on the line $y = 3$, $x = 4$, and point charge $Q = 2 \times 10^{-12}$ C located at origin. Find $\overline{E}$ due to a) Line charge at $P_2(8, 9, 10)$ b) Point charge at $P_2(8, 9, 10)$	Understand	6
C) What are the Applications and Types of Transmission Line?	Remember	6
Q. 3 Solve Any Two of the following.		12
A) Given the following values for $P_1, P_2, I_1 dl_1$ Calculate dH_2 a) $P_1(4, 0, 0)$, $P_2(0, 3, 0)$, $2\pi \overline{az}$ μAm	Understand	6

b) $P_1(4,-2,3)$, $P_2(1,3,2)$, $2\pi(0.6\bar{ax} - 0.8\bar{ay})\mu\text{Am}$

- B)** Show that Magnetic Field Intensity at any point due to finite conductor carrying current I placed along Z axis is **Understand** **6**

$$\bar{H} = \frac{I}{4\rho\pi} (\sin \alpha_2 - \sin \alpha_1) \bar{a}_\phi$$

Where α_2, α_1 are the Inclination of Upper end and Lower end of current carrying conductor.

- C)** A current at 0.4Amp is in $\bar{az}$ direction in the free space in filament parallel to z axis and passing through point $(2, -4, 0)$. Find magnitude of H at $(0, 1, 0)$ if filament lies in the interval $-\infty < Z < \infty$. **Understand** **6**

Q.4 Solve Any Two of the following. **12**

- A)** In the region where $\sigma = 0$, $\epsilon_r = 2.5$, $\mu_r = 10$. Determine whether following pairs of field satisfy Maxwell's equation, $\bar{E} = 2y\bar{ay}$, $\bar{H} = 5x\bar{ax}$ **Understand** **6**
- B)** Define: Propagation constant, characteristic impedance, reflection coefficient and VSWR **Understand** **6**
- C)** If an electric vector A is incident at a boundary between two different dielectric medium with permittivity ϵ_{r1} and ϵ_{r2} at an angle of incidence θ_1 and Let vector B refracted at an angle of refraction θ_2 . Then prove that $\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$. **Understand** **6**

Q. 5 Solve Any Two of the following. **12**

- A)** For Poor conductor, prove that **Understand** **6**
- $$\alpha = \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} \quad \text{and} \quad \beta = \omega \sqrt{\epsilon \mu} \left[1 + \frac{1}{8} \left(\frac{\sigma}{\omega \epsilon} \right)^2 \right]$$
- B)** A 10 GHz plane wave travelling in free space has amplitude of 15 V/m. Find Velocity of propagation, Wavelength, characteristic impedance of medium, amplitude of H , propagation constant. **Understand** **6**
- C)** Explain In short **Remember** **6**
- (i) Biot savart's law
- (ii) Maxwell's Equation in final forms

***** End *****